

RADHEY KALRA

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Open source contributor and full-stack developer working across ML/audio systems, Linux tooling, and self-hosted infrastructure.

EDUCATION

Bachelor of Technology in CSE

Model Institute of Engineering and Technology

Schooling (Class I – XII)

Air Force School, Jammu

Aug. 2023 – May 2027

Jammu and Kashmir, India

2011 – 2023

Jammu and Kashmir, India

PROJECTS

BoxBox

Nov. 2025 – Present

Self-hosted file management system for Linux servers

- Built a lightweight self-hosted file manager in **Go** + **Svelte** for multi-mount browsing, chunked/resumable uploads, live job progress, media preview/streaming, and in-browser editing via **Monaco**.
- Ships as a **31MB single binary** with the UI embedded, plus a published Docker image (**latest**) for one-command homelab/NAS deploys.
- Grew the project to **200+** **GitHub stars** and **5K+** **downloads**.

U-Net Audio Filter

Jul. 2025 – Present

Audio enhancement by removing background noise

- Building an audio enhancement system with a **custom U-Net-inspired architecture** and end-to-end audio denoising pipeline.
- Implemented **Python/PyTorch** + **audio DSP** preprocessing, spectrogram modeling, and evaluation tooling, with early results showing minor gains over open-source baselines.

HPomen Linux (Victus Control)

Mar. 2026 – Jul. 2026

Reverse-engineering HP OMEN/Victus control utility behavior on Linux

- Built a Linux-first OMEN/Victus control suite in **Vala** with a GTK4 monitor app, tray companion, D-Bus helper, and probe CLI.
- Reverse-engineered **HP WMI/sysfs** behavior to validate profile/fan controls (**auto/max**) and extend **hp_wmi** with capability-gated safety fallbacks.
- Submitted a Linux kernel **hp-wmi** patch accepted into **platform-driver-x86**, adding Victus 15-fb0xxx board support for fan-control feature data; also provided **Tested-by** validation for another **hp-wmi** fan-control patch on real hardware.

Fast Blob Storage (FBS)

Apr. 2026 – May 2026

Self-hosted S3-compatible object storage for single-node / homelab deploys

- Built a single-node **S3-compatible** object store in **Go** (filesystem blobs + **SQLite** metadata) with a **SvelteKit** dashboard, shipped as a single binary and Docker image (**latest**).
- Features dual auth (**SigV4** + bearer), multipart uploads, a management API, and zero-copy Linux **sendfile()** egress for fast local downloads without distributed-storage overhead.
- Benchmarks: **~15MB** binary, **~14–16MB** idle RSS, up to **~1.3 GB/s** downloads / **~3.8 GB/s** concurrent aggregate—about **12–14×** less memory than MinIO; **50+** **GitHub stars**.

Additional Projects

- **QckPages**: SvelteKit/Convex app for publishing HTML pages, FBS-backed storage.
- **rsclip**: Rust Wayland clipboard manager with a low-memory daemon, GTK4 UI, SQLite history, search, filters, OCR Support and Secrets management.
- **Dots-Niri**: Replicates my NiriWM rice on Arch-based install with a single **curl** command.
- **HowTo**: Astro tutorials site for practical hardware and software walkthroughs.
- **Pico32**: C++/PlatformIO serial TUI built for ESP32 Wi-Fi and Bluetooth pentesting workflows.
- **TL;DR**: Electron app that summarizes WhatsApp chats with OpenRouter compatible AI models.
- **Hotel Booking**: Svelte/Express/MySQL booking platform with authentication and admin workflows.
- **Tenant Manager**: React Native app for managing tenants and tracking electricity usage.

CO-CURRICULAR ACTIVITIES

Cloud Lead

Aug. 2025 – Present

GDGoC (Google Developer Groups on Campus), MIET Jammu

Jammu and Kashmir, India

- Led cloud learning initiatives through hands-on **Google Cloud** workshops and guided sessions for **100+** **students**, and built a tutorial platform with **20+** lab walkthroughs, notes, and video resources to support **Google Cloud Study Jam** completion.

Cloud and DevOps Lead

Jan. 2026 – Present

AWS Cloud Club MIET Jammu

Jammu and Kashmir, India

- Lead cloud and DevOps initiatives through hands-on workshops, technical sessions, and collaborative projects, driving adoption of **AWS**, containerization, and infrastructure best practices focused on deployment automation and scalability.
- IDE Bootcamp Edition 3 Team Leader – 2nd Place**

Apr. 2026

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

GNDEC, Ludhiana, Punjab
- Led a team to secure **2nd Place** at the AICTE/MoE Innovation Cell bootcamp by applying rapid prototyping and business modeling to develop an AI-powered B2B SaaS product for automated presentation generation from user-uploaded templates and documents.
- IDE Bootcamp Edition 2 Team Leader**

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

IIT Jammu
- Participated in a 5-day immersive bootcamp at IIT Jammu with Wadhvani Foundation, AICTE, and MoE’s Innovation Cell, gaining business modeling, product design, and entrepreneurship experience while leading a team to develop a “Sports Match Finder” app using design thinking and the Lean Canvas model.
- Hackathon Team Leader**

Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025)

MIET, Jammu
- Led multidisciplinary teams in MIET’s internal hackathon and Smart India Hackathon selection rounds in **2024** and **2025**, reaching the **top 9 teams in the college** both years through collaborative problem-solving, rapid prototyping, and real-world solution design.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, Go, Rust, SQL, Bash, PowerShell, QML, Vala
Frameworks & Platforms: SvelteKit, React, Astro, Tailwind CSS, React Native, Bun
Developer Tools: Git, Docker, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform
Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, GCP
Domains: Audio Processing, Linux Servers, Full-Stack Development

CERTIFICATES

Cybersecurity Coursera: Google Cybersecurity; Cisco Cybersecurity Operations; EC-Council Cybersecurity Attack and Defense Fundamentals; Palo Alto Networks Cybersecurity. NPTEL: Ethical Hacking (IIT Kharagpur)	AI NPTEL: Deep Learning (IIT Ropar); Artificial Intelligence for Management. IBM SkillsBuild: AI Agent Architect. Coursera: ServiceNow AI-Powered Workflows
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